SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	**Assessment:**	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	**Total Marks:**	100
CO2 Statement:	*Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.*		
Student Name:	T. Varun Tej	**Reg. No.:**	192511217
Section / Batch:		**Date:**	24/07/2026

Code of Conduct

I ______T. Varun Tej/192511217______________________________ *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ____T. Varun Tej____

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
		GRAND TOTAL:	**100 Marks**

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	**Course Coordinator**	**Program Director**
Signature: ____________	Signature: ____________	Signature: ____________
Date: ________________	Date: ________________	Date: ________________

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q1 Binary Search

5 marks

Identify the error in the following snippet:

```
int binarySearch(int arr[], int n, int key) {
int low = 0, high = n - 1;
while(low < high) {
int mid = (low + high) / 2;
if(arr[mid] == key)
return mid;
else if(arr[mid] < key)
low = mid;
else
high = mid;
}
return -1;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
int binarysearch(int arr[],int n,int key){
int low = 0,high = n-1;
    while(low <= high){
        int mid = (low+high)/2;
        if (arr[mid] == key)
            return mid;
        else if (arr[mid]<key)
            low=mid+1;
        else
            high = mid-1;
    }
    return -1;
}
int main(){
    int arr[]={10,20,30,40,50};
    int n=5;
    int key =30;
    printf("%d",binarysearch(arr,n,key));
    return 0;
}
```

Test Input

stdin input (optional)

Output

2

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q2 **Stack** 5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {
if(top == 0) {
printf("Underflow");
return -1;
}
return stack[top--];
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#define max 100
int stack[max];
int top = -1;
int pop(){
    if (top == -1){
        printf("underflow");
        return -1;
    }
    return stack[top--];
}
int main()
{
    stack[++top] = 10;
    stack[++top] = 20;
    stack[++top] = 30;
    printf("deleted element: %d",pop());
    return 0;
}
```

Test Input

stdin input (optional)

Output

```
deleted element: 30
```

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QUESTIONS

Q1 Binary Search 5 marks
Q2 Stack 5 marks
Q3 Stack 5 marks
Q4 Stack 5 marks
Q5 Stack · Balancing Symbols 5 marks
Q6 Singly Linked List 5 marks
Q7 Stack 5 marks
Q8 Linked List 5 marks
Q9 Linked List 5 marks
Q10 Queue 5 marks

10/10 answered

Q3 **Stack** 5 marks

Check the snippet and identify the logical error.

int stack[5];
int top;
void init() {
top = 0;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#define max 5
int stack[max];
int top;
void init()
{
    top = -1;
}
void push(int value)
{ if(top==max-1)
{
    printf("stack over\n");
    return;
}
 stack[++top] = value;
}
int pop()
{
    if(top == -1)
    {
        printf("stack underflow\n");
        return -1;
    }
    return stack[top--];
}
int main()
{
    init();
    push(10);
    push(20);
    push(30);
    printf("popped element:%d\n",pop());
    return 0;
}
```

Test Input

stdin input (optional)

Output

```
popped element:30
```

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack · Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q4 **Stack** 5 marks

Identify the errors in the following snippet.

```
if(ch == ')') {
while(stack[top] != ')') {
postfix[j++] = pop();
}
pop();
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#include <string.h>

#define MAX 100

char stack[MAX];
int top = -1;
char postfix[MAX];
int j = 0;

char pop()
{
    if (top == -1)
        return '\0';

    return stack[top--];
}

int main()
{
    char ch = ')';

    if (ch == ')')
    {
        while (top != -1 && stack[top] != '(')
        {
            postfix[j++] = pop();
        }

        if (top != -1 && stack[top] == '(')
            pop();
    }

    postfix[j] = '\0';

    printf("Postfix: %s\n", postfix);

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Postfix:
```

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q5 Stack - Balancing Symbols

5 marks

In balancing Bracket, you identify the error.
if((ch == ']' && stack[top] == '[') ||
(ch == '}' && stack[top] == '{')) {
pop();
} else {
return 0;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

#define MAX 100

char stack[MAX];
int top = -1;

void push(char ch)
{
    stack[++top] = ch;
}

char pop()
{
    return stack[top--];
}

int main()
{
    char ch = ']';

    push('[');

    if (top >= 0 &&
        ((ch == ']' && stack[top] == '[') ||
         (ch == '}' && stack[top] == '{') ||
         (ch == ')' && stack[top] == '(')))
    {
        pop();
        printf("Brackets are balanced\n");
    }
    else
    {
        printf("Brackets are not balanced\n");
    }

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Brackets are balanced
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1 Binary Search 5 marks
Q2 Stack 5 marks
Q3 Stack 5 marks
Q4 Stack 5 marks
Q5 Stack - Balancing Symbols 5 marks
Q6 Singly Linked List 5 marks
Q7 Stack 5 marks
Q8 Linked List 5 marks
Q9 Linked List 5 marks
Q10 Queue 5 marks

10/10 answered

Q6 **Singly Linked List** 5 marks

What is the issue?
temp = head
while temp.next != NULL:
print(temp.data)
temp = temp.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <stdlib.h>

struct Node
{
    int data;
    struct Node *next;
};

int main()
{
    struct Node *head = NULL;
    struct Node *temp;

    // Creating nodes
    head = (struct Node *)malloc(sizeof(struct Node));
    head->data = 10;
    head->next = NULL;

    head->next = (struct Node *)malloc(sizeof(struct Node));
    head->next->data = 20;
    head->next->next = NULL;

    head->next->next = (struct Node *)malloc(sizeof(struct Node));
    head->next->next->data = 30;
    head->next->next->next = NULL;

    // Traversing the linked list
    temp = head;

    while (temp != NULL)
    {
        printf("%d ", temp->data);
        temp = temp->next;
    }

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
10 20 30
```

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- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q7 **Stack** 5 marks

Point out the error.
POP(stack):
if top == 0:
print("Underflow")
else:
return stack[top--]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

#define MAX 5

int stack[MAX];
int top = -1;

int pop()
{
    if (top == -1)
    {
        printf("Underflow\n");
        return -1;
    }
    else
    {
        return stack[top--];
    }
}

int main()
{
    stack[++top] = 10;
    stack[++top] = 20;
    stack[++top] = 30;

    printf("Popped element: %d\n", pop());

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Popped element: 30
```

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Simply Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q8 **Linked List** 5 marks

Identify the errors in the following code

while current != NULL:

current.next = prev

prev = current

current = current.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#include <stdlib.h>

struct Node
{
    int data;
    struct Node *next;
};

int main()
{
    struct Node *head, *current, *prev, *next;

    head = (struct Node *)malloc(sizeof(struct Node));
    head->data = 10;

    head->next = (struct Node *)malloc(sizeof(struct Node));
    head->next->data = 20;

    head->next->next = (struct Node *)malloc(sizeof(struct Node));
    head->next->next->data = 30;
    head->next->next->next = NULL;

    current = head;
    prev = NULL;

    while (current != NULL)
    {
        next = current->next;
        current->next = prev;
        prev = current;
        current = next;
    }

    head = prev;

    current = head;

    printf("Reversed linked list: ");

    while (current != NULL)
    {
        printf("%d ", current->data);
        current = current->next;
    }

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Reversed linked list: 30 20 10
```

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- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q9 **Linked List** 5 marks

What is the error in the below code?
new.next = head
head = new

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <stdlib.h>

struct Node
{
    int data;
    struct Node *next;
};

int main()
{
    struct Node *head = NULL;
    struct Node *new;

    new = (struct Node *)malloc(sizeof(struct Node));

    new->data = 10;
    new->next = head;
    head = new;

    printf("Linked List: ");

    struct Node *temp = head;

    while (temp != NULL)
    {
        printf("%d ", temp->data);
        temp = temp->next;
    }

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Linked List: 10
```

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- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q10 Queue

5 marks

What is missing in the below pseudocode?
DEQUEUE(Q):
if front == -1:
print("Underflow")
else:
return Q[front++]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>

#define MAX 5

int Q[MAX];
int front = -1;
int rear = -1;

int dequeue()
{
    int value;

    if (front == -1)
    {
        printf("Underflow\n");
        return -1;
    }
    else
    {
        value = Q[front++];

        if (front > rear)
        {
            front = -1;
            rear = -1;
        }

        return value;
    }
}

int main()
{
    Q[0] = 10;
    Q[1] = 20;
    Q[2] = 30;

    front = 0;
    rear = 2;

    printf("Deleted element: %d\n", dequeue());

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Deleted element: 10
```

Run | Save | Submit Fix

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue · Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack · Postfix Expression 5 marks

10/10 answered

Q1 Stack

5 marks

Point out the error?
PEEK(stack):
return stack[top+1]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

#define MAX 5

int stack[MAX];
int top = -1;

void push(int value)
{
    if (top == MAX - 1)
    {
        printf("Stack Overflow\n");
        return;
    }

    stack[++top] = value;
}

int peek()
{
    if (top == -1)
    {
        printf("Stack Underflow\n");
        return -1;
    }

    return stack[top];
}

int main()
{
    push(10);
    push(20);
    push(30);

    printf("Top element: %d\n", peek());

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Top element: 30
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue · Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack · Postfix Expression 5 marks

10/10 answered

Q1 Stack 5 marks

Point out the error?
PEEK(stack):
return stack[top+1]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

#define MAX 5

int stack[MAX];
int top = -1;

void push(int value)
{
    if (top == MAX - 1)
    {
        printf("Stack Overflow\n");
        return;
    }

    stack[++top] = value;
}

int peek()
{
    if (top == -1)
    {
        printf("Stack Underflow\n");
        return -1;
    }

    return stack[top];
}

int main()
{
    push(10);
    push(20);
    push(30);

    printf("Top element: %d\n", peek());

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Top element: 30
```

Run Save Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q5 Stack

5 marks

Point out the error.
If top == MAX:
Overflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#include <stdbool.h>

#define MAX 5

struct Stack {
    int items[MAX];
    int top;
};

void initialize(struct Stack* s) {
    s->top = -1;
}

bool isFull(struct Stack* s) {
    return s->top == MAX - 1;
}

bool isEmpty(struct Stack* s) {
    return s->top == -1;
}

void push(struct Stack* s, int value) {
    if (isFull(s)) {
        printf("Error: Stack Overflow! Cannot push %d.\n", value);
        return;
    }
    s->top++;
    s->items[s->top] = value;
    printf("Pushed: %d\n", value);
}

int pop(struct Stack* s) {
    if (isEmpty(s)) {
        printf("Error: Stack Underflow! Cannot pop.\n");
        return -1;
    }
    int poppedValue = s->items[s->top];
    s->top--;
    return poppedValue;
}

int main() {
    struct Stack myStack;
    initialize(&myStack);

    push(&myStack, 10);
    push(&myStack, 20);
    push(&myStack, 30);
    push(&myStack, 40);
    push(&myStack, 50);

    push(&myStack, 60);

    printf("Popped: %d\n", pop(&myStack));
    printf("Popped: %d\n", pop(&myStack));

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Pushed: 10
Pushed: 20
Pushed: 30
Pushed: 40
Pushed: 50
Error: Stack Overflow! Cannot
```

Run | Save | Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Q4 Stack

5 marks

Identify the issue in the below pseudocode.
POP():
if top == 0 → Underflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#include <stdbool.h>

#define MAX 5

struct Stack {
    int items[MAX];
    int top;
};

void initialize(struct Stack* s) {
    s->top = -1;
}

bool isFull(struct Stack* s) {
    return s->top == MAX - 1;
}

bool isEmpty(struct Stack* s) {
    return s->top == -1;
}

void push(struct Stack* s, int value) {
    if (isFull(s)) {
        printf("Error: Stack Overflow!\n");
        return;
    }
    s->top++;
    s->items[s->top] = value;
}

int pop(struct Stack* s) {
    if (isEmpty(s)) {
        printf("Error: Stack Underflow!\n");
        return -1;
    }
    int poppedValue = s->items[s->top];
    s->top--;
    return poppedValue;
}

int main() {
    struct Stack myStack;
    initialize(&myStack);

    push(&myStack, 10);

    printf("Popped: %d\n", pop(&myStack));
    printf("Popped: %d\n", pop(&myStack));

    return 0;
}
```

Test Input

stdin input (optional)

Output

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QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Dequeue 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q6 **Queue** 5 marks

Identify the error in the Queue enqueue logic.

```
void enqueue(int data) {
if (rear == MAX) return;
queue[rear++] = data;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#include <stdbool.h>

#define max 5

struct Queue {
    int items[max];
    int front;
    int rear;
    int size;
};

void initialize(struct Queue* q) {
    q->front = 0;
    q->rear = 0;
    q->size = 0;
}

bool isFull(struct Queue* q) {
    return q->size == max;
}

bool isEmpty(struct Queue* q) {
    return q->size == 0;
}

void enqueue(struct Queue* q, int data) {
    if (isFull(q)) {
        printf("Error: Queue Overflow!\n");
        return;
    }
    q->items[q->rear] = data;
    q->rear = (q->rear + 1) % max;
    q->size++;
    printf("Enqueued: %d\n", data);
}

int dequeue(struct Queue* q) {
    if (isEmpty(q)) {
        printf("Error: Queue Underflow!\n");
        return -1;
    }
    int data = q->items[q->front];
    q->front = (q->front + 1) % max;
    q->size--;
    return data;
}

int main() {
    struct Queue q;
    initialize(&q);

    enqueue(&q, 10);
    enqueue(&q, 20);
    enqueue(&q, 30);
    enqueue(&q, 40);
    enqueue(&q, 50);

    printf("Dequeued: %d\n", dequeue(&q));
    printf("Dequeued: %d\n", dequeue(&q));

    printf("Testing reuse of space:\n");
    enqueue(&q, 60);
    enqueue(&q, 70);

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Enqueued: 10
Enqueued: 20
Enqueued: 30
Enqueued: 40
Enqueued: 50
```

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QUESTIONS

Q1 Stack 5 marks

Q2 Circular Queue 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Doubly Linked List 5 marks

Q6 Queue 5 marks

Q7 Queue - Deque 5 marks

Q8 Binary Search 5 marks

Q9 Circular Queue 5 marks

Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q6 **Queue** 5 marks

Identify the error in the Queue enqueue logic.

void enqueue(int data) {

if (rear == MAX) return;

queue[rear++] = data;

}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#include <stdbool.h>

#define max 5

struct Queue {
    int items[max];
    int front;
    int rear;
    int size;
};

void initialize(struct Queue* q) {
    q->front = 0;
    q->rear = 0;
    q->size = 0;
}

bool isFull(struct Queue* q) {
    return q->size == max;
}

bool isEmpty(struct Queue* q) {
    return q->size == 0;
}

void enqueue(struct Queue* q, int data) {
    if (isFull(q)) {
        printf("Error: Queue Overflow!\n");
        return;
    }
    q->items[q->rear] = data;
    q->rear = (q->rear + 1) % max;
    q->size++;
    printf("Enqueued: %d\n", data);
}

int dequeue(struct Queue* q) {
    if (isEmpty(q)) {
        printf("Error: Queue Underflow!\n");
        return -1;
    }
    int data = q->items[q->front];
    q->front = (q->front + 1) % max;
    q->size--;
    return data;
}

int main() {
    struct Queue q;
    initialize(&q);

    enqueue(&q, 10);
    enqueue(&q, 20);
    enqueue(&q, 30);
    enqueue(&q, 40);
    enqueue(&q, 50);

    printf("Dequeued: %d\n", dequeue(&q));
    printf("Dequeued: %d\n", dequeue(&q));

    printf("Testing reuse of space:\n");
    enqueue(&q, 60);
    enqueue(&q, 70);

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Enqueued: 10
Enqueued: 20
Enqueued: 30
Enqueued: 40
Enqueued: 50
```

Run Save Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

QUESTIONS

Q1 Stack 5 marks
Q2 Circular Queue 5 marks
Q3 Stack 5 marks
Q4 Stack 5 marks
Q5 Doubly Linked List 5 marks
Q6 Queue 5 marks
Q7 Queue - Deque 5 marks
Q8 Binary Search 5 marks
Q9 Circular Queue 5 marks
Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q7 Queue - Deque

5 marks

Find the error in this Deque insertFront operation.
void insertFront(int data) {
if (front == 0) front = MAX - 1;
else front = front - 1;
deque[front] = data;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <stdbool.h>

#define MAX 5

struct Deque {
    int items[MAX];
    int front;
    int rear;
    int size;
};

void initialize(struct Deque* dq) {
    dq->front = -1;
    dq->rear = -1;
    dq->size = 0;
}

bool isFull(struct Deque* dq) {
    return dq->size == MAX;
}

bool isEmpty(struct Deque* dq) {
    return dq->size == 0;
}

void insertFront(struct Deque* dq, int data) {
    if (isFull(dq)) {
        printf("Error: Deque Overflow! Cannot insert %d at front.\n", d
        return;
    }

    if (isEmpty(dq)) {
        dq->front = 0;
        dq->rear = 0;
    } else if (dq->front == 0) {
        dq->front = MAX - 1;
    } else {
        dq->front = dq->front - 1;
    }

    dq->items[dq->front] = data;
    dq->size++;
    printf("Inserted at front: %d\n", data);
}

void insertRear(struct Deque* dq, int data) {
    if (isFull(dq)) {
        printf("Error: Deque Overflow!\n");
        return;
    }

    if (isEmpty(dq)) {
        dq->front = 0;
        dq->rear = 0;
    } else if (dq->rear == MAX - 1) {
        dq->rear = 0;
    } else {
        dq->rear = dq->rear + 1;
    }

    dq->items[dq->rear] = data;
    dq->size++;
}

int main() {
    struct Deque dq;
    initialize(&dq);
```

Test Input

stdin input (optional)

Output

```
Inserted at front: 10
Inserted at front: 20
Inserted at front: 30

--- Testing Overflow Condition
---
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue · Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack · Postfix Expression 5 marks

10/10 answered

Q8 **Binary Search** 5 marks

Identify the bug in this Recursive Binary Search.
int binarySearch(int arr[], int l, int r, int x) {
int mid = l + (r - l) / 2;
if (arr[mid] == x) return mid;
if (arr[mid] > x) return binarySearch(arr, l, mid, x);
return binarySearch(arr, mid, r, x);
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

int binarySearch(int arr[], int l, int r, int x) {
    if (l > r) {
        return -1;
    }

    int mid = l + (r - l) / 2;

    if (arr[mid] == x) {
        return mid;
    }

    if (arr[mid] > x) {
        return binarySearch(arr, l, mid - 1, x);
    }

    return binarySearch(arr, mid + 1, r, x);
}

int main() {
    int arr[] = {2, 3, 4, 10, 40};
    int n = sizeof(arr) / sizeof(arr[0]);

    int target1 = 10;
    int result1 = binarySearch(arr, 0, n - 1, target1);
    printf("Searching for %d... Found at index: %d\n", target1, result1

    int target2 = 5;
    int result2 = binarySearch(arr, 0, n - 1, target2);
    printf("Searching for %d... Found at index: %d\n", target2, result2

    return 0;
}
```

Test Input

```
stdin input (optional)
```

Output

```
Searching for 10... Found at index:
3
Searching for 5... Found at index:
-1
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

- **Q1** Stack 5 marks
- **Q2** Circular Queue 5 marks
- **Q3** Stack 5 marks
- **Q4** Stack 5 marks
- **Q5** Doubly Linked List 5 marks
- **Q6** Queue 5 marks
- **Q7** Queue · Deque 5 marks
- **Q8** Binary Search 5 marks
- **Q9** Circular Queue 5 marks
- **Q10** Stack · Postfix Expression 5 marks

10/10 answered

Q9 **Circular Queue** 5 marks

Identify the bug in this Circular Queue dequeue logic.
int dequeue() {
if (front == -1) return -1;
int data = queue[front];
front = (front + 1) % MAX;
return data;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#include <stdbool.h>

#define MAX 5

struct CircularQueue {
    int items[MAX];
    int front;
    int rear;
};

void initialize(struct CircularQueue* q) {
    q->front = -1;
    q->rear = -1;
}

bool isFull(struct CircularQueue* q) {
    return ((q->rear + 1) % MAX) == q->front;
}

bool isEmpty(struct CircularQueue* q) {
    return q->front == -1;
}

void enqueue(struct CircularQueue* q, int value) {
    if (isFull(q)) {
        printf("Error: Queue Overflow!\n");
        return;
    }
    if (isEmpty(q)) {
        q->front = 0;
    }
    q->rear = (q->rear + 1) % MAX;
    q->items[q->rear] = value;
    printf("Enqueued: %d\n", value);
}

int dequeue(struct CircularQueue* q) {
    if (isEmpty(q)) {
        printf("Error: Queue Underflow!\n");
        return -1;
    }
```

Test Input

stdin input (optional)

Output

```
Enqueued: 10
Enqueued: 20
Popped: 10
Popped: 20
Testing enqueue after
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q10 **Stack - Postfix Expression** 5 marks

Find the bug in the evaluation of a Postfix expression.

```
while (expression[i] != '\0') {
if (isdigit(expression[i])) push(expression[i] - '0');
else {
int op1 = pop();
int op2 = pop();
// perform operation: result = op1 + op2;
push(result);
}
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#include <ctype.h>
#include <stdlib.h>

#define MAX 100

struct Stack {
    int items[MAX];
    int top;
};

void initialize(struct Stack* s) {
    s->top = -1;
}

void push(struct Stack* s, int value) {
    if (s->top == MAX - 1) return;
    s->items[++(s->top)] = value;
}

int pop(struct Stack* s) {
    if (s->top == -1) return 0;
    return s->items[(s->top)--];
}

int evaluatePostfix(char* expression) {
    struct Stack s;
    initialize(&s);
    int i = 0;

    while (expression[i] != '\0') {
        if (isdigit(expression[i])) {
            push(&s, expression[i] - '0');
        } else {
            int op2 = pop(&s);
            int op1 = pop(&s);
            int result = 0;

```

Test Input

stdin input (optional)

Output

```
Postfix Expression: 231*+9-
Evaluation Result: -4
```